Survival Garden Regional Plans

Complete food production plans for 5 US regions — two growing systems

$18

Open-pollinated · Community-grown · Food sovereignty for everyone

Survival Food Gardens: Five Regional Plans

Condensed Growing for Maximum Calories on Minimal Land

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This is a working document, not a philosophy piece. Five regions, two growing systems each, full seed lists, planting calendars, and layout schematics. The goal is clear: produce as much food as possible from as little ground as possible, in the specific soil and climate you actually live in.

Important Notice

This guide is educational content based on real growing data and regional agricultural experience. Caloric output estimates are based on optimal conditions and will vary with your soil, weather, skill level, and growing season. It is not professional agricultural advice, and no garden plan should be treated as a guaranteed food supply without multiple seasons of proven results in your specific location. Start with the plan, adjust based on observation, and build your knowledge over time.

How to Use This Guide

Each regional plan is self-contained. Find your region. Choose your system. Follow the plan.

System 1 — Minimal Improvement: Works with native soil. Mounded rows or hills, no imported amendments, no permanent structures. This is what you can do right now with a shovel and seeds. Soil is loosened and mounded 6–12 inches above surrounding grade to improve drainage and workability. Mulch (straw, leaves, wood chips) is the only addition. Crop selection is adjusted to suit native soil conditions.

System 2 — Full Improvement: Raised beds, amended soil, season extension. This is what you build over one to two seasons to significantly increase yield and expand crop options. Beds are built 8–18 inches high depending on crop, filled with a mix of native soil, compost, and amendments. Row covers and cold frames extend seasons where applicable.

Plot Size Reference

All plans are designed for a **20' × 40' = 800 square foot** reference plot. This is a realistic backyard footprint. Caloric output estimates are based on this size; scale the plan proportionally for larger or smaller spaces.

Caloric context: An adult needs approximately 700,000–730,000 calories per year. An intensively managed 800 sq ft garden, optimized for caloric crops (potatoes, sweet potatoes, dry beans, winter squash), can realistically produce 300,000–500,000 calories per year — 40–70% of one person's annual needs. With a second 800 sq ft plot, or by extending the season, full caloric self-sufficiency approaches.

How to Read the Schematics

All schematics show a bird's-eye view of the 20' × 40' plot. North is at the top of every diagram.

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Each grid cell = 2' × 2' (4 sq ft)
10 columns across × 20 rows deep = 800 sq ft
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Two-letter crop codes used throughout:

Code	Crop	Code	Crop
3S	Three Sisters zone (corn/bean/squash)	GR	Mixed greens (collards/kale/mustard)
SW	Sweet Potato	GA	Garlic
OK	Okra	ON	Onion
CP	Cowpea / Southern Pea	CA	Carrot
KL	Kale / Collards	BT	Beet
TM	Tomato	TN	Turnip / Rutabaga
PT	Potato	PZ	Peas
WS	Winter Squash / Pumpkin	SU	Sunflower
DB	Dry Beans	HB	Herbs (basil/dill/parsley)
PK	Pickle Cucumber	PP	Pepper
EG	Eggplant	RD	Radish
WM	Watermelon / Melon	MG	Microgreens / Fast greens
--	Walking path (2' wide)	..	Border (garlic/herb/allium)

Orientation note: In all northern regions (Wisconsin, Michigan, Arkansas), tall crops (corn, sunflowers, trellised tomatoes) are placed on the NORTH side of the plot so they don't shade shorter crops. In Houston, where summer shade is beneficial, tall crops may be positioned on the WEST side to provide afternoon shade to heat-sensitive plants.

Companion Planting Guilds

These plant relationships appear throughout the regional plans. Understand why they're placed together.

Three Sisters (Corn + Climbing Beans + Winter Squash)
The oldest companion system in North America. Corn provides structure for beans to climb. Beans fix atmospheric nitrogen and feed it to corn's heavy appetite. Squash sprawls across the ground, shading weeds, retaining soil moisture, and deterring pests with its rough leaves. Together, they produce three distinct caloric foods from the same square footage. Plant corn first in hills, add beans when corn is 6–8" tall, add squash between hills two weeks after beans.

Allium Border (Garlic + Onion on Bed Edges)
Garlic and onions planted along the outer edges of beds emit sulfur compounds that deter aphids, spider mites, Japanese beetles, and several soil pests. They also repel rabbits and deer. Plant garlic along all four edges of any bed — it costs almost nothing in space and pays in pest suppression across the entire bed.

Tomato Guild (Tomato + Basil + Carrot)
Basil planted beneath tomatoes repels aphids and whiteflies and may improve tomato fruit flavor. Carrots loosen compacted soil around tomato roots. Keep the three within 2–3 feet of each other. Do not plant basil on the shaded north side of tomatoes — it needs sun.

Brassica Guild (Brassicas + Dill + Alliums)
Dill and fennel attract parasitic wasps that prey on caterpillars (cabbageworm, cabbage looper). Plant dill every 4 feet along brassica rows. Garlic and onions in adjacent rows deter aphids. Radishes act as trap crops for flea beetles — plant them at row ends.

Root Companionship (Carrot + Leek/Onion)
Carrots and onions confuse each other's primary pests. Carrot fly is deterred by onion smell; onion fly is deterred by carrot smell. Interplant them in the same rows, alternating.

Anti-Companions — What Not to Plant Together

- Onions and garlic near beans or peas (stunts legume growth significantly)
- Fennel near almost anything (releases allelopathic compounds that inhibit most vegetables — keep isolated or skip entirely)
- Brassicas near strawberries (mutual growth suppression)
- Sweet potatoes near winter squash (both sprawl aggressively — they will compete for ground)
- Potatoes near tomatoes (share diseases, particularly late blight — keep separated and rotate together)

HOUSTON, TEXAS

Harris County / Gulf Coast Prairie · Zone 9a–9b · 290+ Frost-Free Days

Houston Climate Profile

Houston's growing season is more than nine months long, but the most productive windows are fall (mid-August through December) and spring (mid-February through early June). Summer — June through mid-August — is brutal: 95–102°F with heat indexes exceeding 110°F and humidity over 80%. Most vegetables either slow drastically or die outright. The crops that thrive in Houston summer are the ones worth knowing.

The soil is the main challenge. Houston's native soil is Houston Black Clay — a Vertisol that swells when wet into a near-impenetrable mass, then cracks into deep fissures when dry. Roots struggle. Drainage is poor. Without any improvement, you must mound aggressively.

Factor	Data
Zone	9a (northern Houston) to 9b (south)
Last Spring Frost	February 10–20
First Fall Frost	December 1–10
Frost-Free Days	290–310
Annual Rainfall	49–52 inches (relatively even distributi
Summer Avg High	94–97°F (heat index 105–112°F)
Winter Avg Low	38–44°F (occasional hard freezes to 25°F
Native Soil	Houston Black Clay (Vertisol) — expansiv
Soil pH	7.5–8.2 (alkaline) — important for nutri

Three Growing Windows:

1. Spring Season: February 20 – June 10
2. Summer: June 10 – August 15 (heat crops only: okra, cowpeas, peppers, sweet potatoes)
3. Fall Season: August 15 – December 10 (the most productive season)

Houston — Seed List

Quantities for 800 sq ft total. Adjust to your actual planting area.

Crop	Best Variety	Days to Maturity	Qty (800 sq ft)	Best Season	Purpose
Sweet Potato	Beauregard, Covington	90–120 days	50–75 slips	Summer	Primary calories
Cowpea / Southern Pea	Pinkeye Purple Hull, Iron & Clay	60–70 days	½ lb	Summer/Fall	Protein + nitrogen
Okra	Clemson Spineless, Burgundy	50–60 days	1 oz	Summer	Calories + vitamins
Collard Greens	Champion, Georgia Southern	60–75 days	1 pkt	Fall/Winter	Vitamins A, C, K
Mustard Greens	Florida Broadleaf, Southern Giant Curled	40–50 days	1 pkt	Fall/Spring	Fast vitamin crop
Kale	Vates, Lacinato	50–60 days	1 pkt	Fall/Winter	Vitamins, frost-hardy
Tomato (spring)	Celebrity, Solar Fire, Heatmaster	70–80 days	1 pkt (10 plants)	Spring	Vitamins, lycopene
Tomato (fall)	Sweet 100 Cherry, Juliet	60–65 days	1 pkt (10 plants)	Fall	Shorter season
Pepper	Carmen, Jimmy Nardello, Cayenne	70–80 days	1 pkt (8 plants)	Spring/Fall	Vitamins, storage (dried)
Eggplant	Black Beauty, Ichiban	65–80 days	1 pkt (6 plants)	Spring/Summer	Heat-tolerant calories
Garlic	Creole varieties (Ajo Rojo, Creole Red)	8–9 months	2 lbs (seed)	Plant Oct, harvest May	Flavor, pest deterrent
Armenian Cucumber	Armenian (not true cucumber)	60–70 days	1 pkt	Spring/Summer	Heat-tolerant; high production
Watermelon	Sugar Baby, Jubilee	75–90 days	1 pkt	Spring	Calories, water content
Sunflower	Mammoth Russian	80–90 days	½ oz	Spring	Seeds (fat/protein), trellis
Basil	Italian Large Leaf, Thai	60 days	1 pkt	Spring/Summer	Tomato companion, culinary
Dill	Bouquet	40–50 days	1 pkt	Fall	Brassica companion
Radish	Cherry Belle, Daikon	25–45 days	1 pkt	Fall/Spring	Fast succession, trap crop
Turnip	Purple Top White Globe, Hakurei	40–55 days	1 pkt	Fall/Winter	Roots + greens
Beans (snap)	Rattlesnake Pole, Kentucky Wonder	55–65 days	¼ lb	Spring/Fall	Protein
Dry Beans	Pinquito, Black Turtle	90 days	¼ lb	Spring (dry season)	Storage protein

Note on sweet potatoes: These are grown from slips (rooted cuttings), not seeds. Purchase slips from a reputable nursery or produce your own by burying a sweet potato halfway in water in early spring. Slips root from the submerged portion and are separated and planted when 6–8" long.

Houston — Planting Calendar

Month	Planting Activity
January	Start tomato and pepper seeds indoors. P
February 15	Direct sow radishes, turnips, mustard gr
February 20–28	Set out tomato and pepper transplants af
March	Direct sow cowpeas (spring planting), cu
April	Prime spring production. Harvest tomatoe
May	Harvest spring crops aggressively. Start
June 1–15	Transition to summer mode. Plant sweet p
June–August	Summer mode: Harvest okra daily (it toug
August 15	Begin fall season. Direct sow collards,
September 1	Set out fall tomato and pepper transplan
October	Plant garlic (main annual planting). Pri
October–December	Harvest collards, kale, mustard (frost i
November–December	Harvest sweet potatoes. Late fall tomato
December–January	Plant more cool-season greens. Begin pla

Houston — System 1: Minimal Improvement (Mounded Rows)

What "minimal" means in Houston clay: Mound soil 10–14 inches above surrounding grade using only native soil. Do not import amendments. Work the native clay into ridges and allow them to dry slightly between rains for a firmer planting surface. Plant into the top of the mounds. The mounding improves drainage enough to prevent root rot and creates aerated soil for root crops.

Row orientation: Run rows east-west in spring/summer to use tall crops on the WEST and SOUTH for afternoon shade of heat-sensitive plants. In fall, run rows north-south to maximize sun on all plants.

Crop adjustments for raw clay: Skip carrots (clay causes forking), skip parsnips, skip standard potatoes (poorly draining clay causes rot). Focus on: sweet potatoes (clay-tolerant when mounded), cowpeas (clay-tolerant legumes), okra (thrives in clay), collards/kale (clay-tolerant), eggplant, peppers.

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HOUSTON — SYSTEM 1 LAYOUT (Mounded Rows in Clay)
Plot: 20' Wide × 40' Long · Each cell = 2'×2' · N at top

FALL/SPRING LAYOUT (shown — summer layout note below)

     +------------ 20 FEET ------------>
              NORTH
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+----+----+----+----+----+----+----+----+----+----+
- GA - KL - KL - KL - KL - KL - KL - KL - KL - GA - ← 2'   Row 1
+----+----+----+----+----+----+----+----+----+----+
- GA - KL - KL - KL - KL - KL - KL - KL - KL - GA - ← 4'   Row 2  [COLLARD/KALE BED - 8'x16']
+----+----+----+----+----+----+----+----+----+----+
- GA - TN - TN - TN - TN - TN - TN - TN - TN - GA - ← 6'   Row 3  [TURNIPS + GARLIC BORDER]
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 8'   PATH
+----+----+----+----+----+----+----+----+----+----+
- ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 10'  Row 5  [TOMATO + BASIL + ONION]
+----+----+----+----+----+----+----+----+----+----+
- ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 12'  Row 6
+----+----+----+----+----+----+----+----+----+----+
- ON - PP - PP - EG - PP - PP - EG - PP - PP - ON - ← 14'  Row 7  [PEPPERS + EGGPLANT]
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 16'  PATH
+----+----+----+----+----+----+----+----+----+----+
- SU - SW - SW - SW - SW - SW - SW - SW - SW - SU - ← 18'  Row 9  [SWEET POTATO (main block)]
+----+----+----+----+----+----+----+----+----+----+
- SU - SW - SW - SW - SW - SW - SW - SW - SW - SU - ← 20'  Row 10
+----+----+----+----+----+----+----+----+----+----+
- SU - SW - SW - SW - SW - SW - SW - SW - SW - SU - ← 22'  Row 11 [SWEET POTATO cont'd]
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 24'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - CP - CP - CP - CP - CP - CP - CP - CP - GA - ← 26'  Row 13 [COWPEA/SOUTHERN PEA]
+----+----+----+----+----+----+----+----+----+----+
- GA - CP - CP - CP - CP - CP - CP - CP - CP - GA - ← 28'  Row 14
+----+----+----+----+----+----+----+----+----+----+
- GA - OK - OK - OK - OK - OK - OK - OK - OK - GA - ← 30'  Row 15 [OKRA - South end, full sun]
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 32'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - GR - GR - GR - GR - GR - GR - GR - GR - GA - ← 34'  Row 17 [MUSTARD/GREENS rotation]
+----+----+----+----+----+----+----+----+----+----+
- GA - GR - GR - GR - GR - GR - GR - GR - GR - GA - ← 36'  Row 18
+----+----+----+----+----+----+----+----+----+----+
- GA - RD - TN - RD - TN - RD - TN - RD - TN - GA - ← 38'  Row 19 [RADISH/TURNIP alternating]
+----+----+----+----+----+----+----+----+----+----+
- GA - RD - TN - RD - TN - RD - TN - RD - TN - GA - ← 40'  Row 20
+----+----+----+----+----+----+----+----+----+----+
                    SOUTH
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LEGEND: GA=Garlic border, KL=Kale/Collards, TN=Turnips, TM=Tomato, HB=Herbs/Basil,
ON=Onion, PP=Pepper, EG=Eggplant, SU=Sunflower, SW=Sweet Potato,
CP=Cowpea/Southern Pea, OK=Okra, GR=Greens (mustard/mizuna), RD=Radish, --=Path

SUMMER SWAP: Replace Rows 1-7 with full SW (sweet potato) and OK (okra) blocks.

Move CP to Rows 13-14. Tomatoes/peppers removed until Aug 15 replanting.

Companion rationale:

- Garlic borders every zone — pest deterrent across the entire plot
- Sunflowers flanking the sweet potato block — provide partial shade for vines, seeds as food
- Okra on the south end — gets maximum sun, doesn't shade others
- Tomato/basil paired in center — convenient harvest, mutual benefit
- Rotation: Sweet potatoes go where cowpeas were the prior year (nitrogen benefit)

Houston — System 2: Full Improvement (Raised Beds)

What changes with improvements:

- Three raised beds, 4' wide each, built 12–18" above grade
- Soil mix: 40% topsoil, 40% compost, 20% coarse pine bark or perlite (critical for drainage past the clay base)
- Drip irrigation or soaker hose runs through each bed
- Now possible: Root crops (carrots, beets, radishes) grow well in amended, loose raised bed soil
- Shade cloth (30–40%) over tomato/pepper beds in June–August extends summer productivity
- Drip system enables precision watering through dry spells

New crops enabled by System 2:
Add carrots (Danvers 126 — heat-tolerant), beets (Merlin, Chioggia), Armenian cucumber directly (better drainage needed), dry beans (requires well-drained soil).

Layout: Same general zone placement as System 1, but organized into three permanent 4' beds running north-south with 2' paths between. South end (full sun, hottest zone) becomes the permanent sweet potato and okra section. North section rotates between cool-season crops. Middle section is the permanent tomato-pepper-eggplant bed.

Season extension: A 30% shade cloth over the middle bed in June–August allows tomatoes and peppers to continue producing through summer heat — normally not possible in Houston without this intervention. This can add a full summer season of production.

Houston — Caloric Output Estimate

Crop	Area (sq ft)	Est. Yield	Calories
Sweet Potato	160 sq ft	100–120 lbs	42,000–50,000
Cowpeas (dry)	64 sq ft	8–10 lbs	12,000–15,000
Okra	48 sq ft	30–40 lbs fresh	4,000–5,000
Tomatoes	80 sq ft	60–80 lbs	9,000–12,000
Collards/Kale	80 sq ft	40–60 lbs	6,000–9,000
Peppers + Eggplant	48 sq ft	20–30 lbs	3,000–5,000

Greens/Turnips	80 sq ft	30–50 lbs combined	4,000–7,000
TOTAL (approx)	560 sq ft active		~80,000–103,000 cal/season

Houston's three seasons allow this plot to produce approximately 200,000–250,000 calories per year — roughly 25–35% of one adult's annual caloric needs from 800 sq ft.

NORTHWEST ARKANSAS

Washington & Benton Counties · Zone 7a · Ozark Plateau · 165–219 Frost-Free Days

Northwest Arkansas Climate Profile

Northwest Arkansas is rich growing country with two productive seasons and one brutal summer between them. The Ozark Plateau brings rocky, shallow soils — frequently only 8–18 inches before hitting chert-rich clay or limestone — and a two-season gardening rhythm that experienced growers here have worked with for generations. Spring rains can be intense; summer heat and humidity favor disease; fall is often the best season.

Factor	Data
Zone	7a (with 6b pockets at higher elevation)
Last Spring Frost	April 16 average (50% risk April 23; ext
First Fall Frost	October 7 risk begins (50% by October 24
Frost-Free Days	165–219 (Bentonville ~165; Fayetteville
Annual Rainfall	45–48 inches (spring heaviest, particula
Summer Avg High	86–89°F (heat index 105°F+ in July due t
Winter Lows	25°F average; ice storms and dramatic sw
Native Soil	Rocky Clarksville cherty silt loam / cla
Soil pH	5.0–6.5 (often acidic — good for potatoe

Key regional challenge: Summer disease pressure. Late blight on tomatoes and potatoes; downy mildew on cucumbers; fungal diseases on beans. Variety selection is not optional — it's the difference between harvest and crop failure.

Arkansas-developed varieties: The University of Arkansas breeding program has produced regionally optimized varieties worth prioritizing — Arkansas Traveler tomato, Ozark Pink tomato, and the Beauregard sweet potato were all developed here for these specific conditions.

Northwest Arkansas — Seed List

Crop	Best Variety	Days	Qty (800 sq ft)	Season	Purpose
Potato	Elba (late blight resistance), Yukon Gem	70–90 days	5 lbs (seed)	Spring (Feb–Mar)	Primary calories
Sweet Potato	Beauregard, Covington	90–120 days	40–50 slips	Summer start	Primary calories
Dry Beans	Black Turtle, Appalachian October	85–95 days	½ lb	Spring/Summer	Storage protein
Cowpea	Pinkeye Purple Hull	60–70 days	¼ lb	Summer	Protein, heat-tolerant
Winter Squash	Waltham Butternut, Jarrahdale	85–100 days	1 pkt (8 plants)	Spring plant, fall harvest	Long-storage calories
Tomato (spring)	Arkansas Traveler, Ozark Pink	75–85 days	1 pkt (10 plants)	Spring	Disease-resistant, adapted
Tomato (fall)	Celebrity, Sweet Million Cherry	70 days	1 pkt (8 plants)	Fall (plant Aug)	Short fall window
Pepper	Jimmy Nardello, Carmen, Cayenne	65–75 days	1 pkt (8 plants)	Spring/Fall	Storage (dried), vitamins
Corn	Bloody Butcher (dent, drying), Hopi Blue	100–110 days	¼ lb	Spring plant	Dried cornmeal, storage
Collard Greens	Georgia Southern, Champion	60–75 days	1 pkt	Fall/Winter	Vitamins, frost-hardy
Kale	Vates, Winterbor	50–60 days	1 pkt	Spring/Fall	Cold-hardy vitamins
Garlic	German Extra Hardy, Chesnok Red	8–9 months	2 lbs	Plant Oct, harvest June	Pest deterrent, culinary
Onion	Candy, Red Creole	100–120 days	1 pkt or transplants	Spring	Companion, calories
Carrot	Danvers 126, Bolero	70–80 days	1 pkt	Spring/Fall	Vitamins, root cellar
Beet	Detroit Dark Red, Chioggia	55–65 days	1 pkt	Spring/Fall	Root + greens
Turnip	Purple Top White Globe, Hakurei	40–55 days	1 pkt	Fall (main)	Fast fall crop
Radish	Cherry Belle, Daikon	25–60 days	1 pkt	Spring/Fall	Trap crop + fast food
Snap Beans	Provider (disease-resistant), Dragon Ton	50–55 days	¼ lb	Spring/Fall	Fresh + dry
Cucumber	Spacemaster (powdery mildew tolerant)	55–60 days	1 pkt	Spring	Disease-resistant required
Sunflower	Mammoth Russian, Peredovik (soil type)	80–90 days	½ oz	Spring	Seed fat/protein, bird control
Basil	Genovese, Siam Queen	60 days	1 pkt	Spring	Tomato companion
Dill	Bouquet	40–50 days	1 pkt	Spring/Fall	Brassica companion
Peas	Wando, Oregon Sugar Pod	60–70 days	¼ lb	Early spring	Spring protein

Northwest Arkansas — Planting Calendar

Month	Activity
Feb (late)	Direct sow peas. Prepare beds if soil is
Mar 1–15	Plant seed potatoes. Direct sow onion se
Mar 15 – Apr 1	Start tomato, pepper transplants indoors
Apr 1–15	Sow snap beans (can risk light frost wit
Apr 23–30	LAST FROST RISK PAST. Set out tomato, pe
May 1–15	Plant sweet potato slips when soil reach
May – June	Heavy spring rains — ensure drainage. Mo
June 15 – July	Transition to heat mode. Harvest potatoe
July 15 – Aug 1	Critical fall planting window. Direct so
August 1–15	Set out fall tomato, pepper transplants
Sept – Oct	Prime fall production. Harvest regularly

Nov	Dig sweet potatoes before hard freeze. H
Dec – Jan	Mulch garlic heavily. Plan next season.

Northwest Arkansas — System 1: Minimal Improvement (Mounded Rows)

What "minimal" means on Ozark soils: The rocky, shallow nature of Ozark soils requires working with what you have. In deeper clay areas, mound 8–10 inches above grade, same as Houston. In very rocky areas, identify the least rocky pockets and build rows there. Remove surface rocks and use them to build low retaining walls along row edges — this is free infrastructure that improves the rows over time. Skip root crops where rock density is high; focus on surface-rooting crops (sweet potatoes, beans, squash, greens).

Disease management in System 1: Without raised beds and amended soil, air circulation between plants is the primary disease tool. Wider spacing than you think necessary, stake/trellis everything possible, and mulch heavily to prevent soil splash.

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NW ARKANSAS — SYSTEM 1 LAYOUT (Mounded Rows, Ozark Soil)
Plot: 20' Wide × 40' Long · Each cell = 2'×2' · N at top

SPRING LAYOUT (main planting shown)

      +------------ 20 FEET ------------>
                NORTH
  +----+----+----+----+----+----+----+----+----+----+
  - SU - SU - SU - SU - SU - SU - SU - SU - SU - SU - ← 2'   [SUNFLOWERS - N edge, windbreak + seeds]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 4'   [POTATOES - Row 1]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 6'   [POTATOES - Row 2]
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 8'   PATH
  +----+----+----+----+----+----+----+----+----+----+
  - ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 10'  [TOMATO + BASIL + ONION]
  +----+----+----+----+----+----+----+----+----+----+
  - ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 12'
  +----+----+----+----+----+----+----+----+----+----+
  - ON - PP - PP - PP - PP - PP - PP - PP - PP - ON - ← 14'  [PEPPERS]
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 16'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 18'  [THREE SISTERS - corn/bean/squash]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 20'
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 22'
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+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - - -- - -- - -- - - ← 24'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - DB - DB - CA - CA - CA - CA - DB - DB - GA - ← 26'  [DRY BEANS + CARROTS interplanted]
+----+----+----+----+----+----+----+----+----+----+
- GA - DB - DB - CA - CA - CA - CA - DB - DB - GA - ← 28'
+----+----+----+----+----+----+----+----+----+----+
- GA - SW - SW - SW - SW - SW - SW - SW - SW - GA - ← 30'  [SWEET POTATO - mounded rows]
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - - -- - -- - -- - - ← 32'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - KL - KL - KL - BT - BT - KL - KL - KL - GA - ← 34'  [KALE + BEETS - fall crop zone]
+----+----+----+----+----+----+----+----+----+----+
- GA - KL - KL - KL - BT - BT - KL - KL - KL - GA - ← 36'
+----+----+----+----+----+----+----+----+----+----+
- GA - TN - TN - RD - TN - TN - RD - TN - TN - GA - ← 38'  [TURNIPS + RADISH]
+----+----+----+----+----+----+----+----+----+----+
- GA - TN - TN - RD - TN - TN - RD - TN - TN - GA - ← 40'
+----+----+----+----+----+----+----+----+----+----+
                    SOUTH

FALL SWAP: After potatoes are pulled (June), Rows 4-6 become second Three Sisters planting
OR a full fall brassica/greens block. Rotate potatoes away from this zone next year.
```

Northwest Arkansas — System 2: Full Improvement

What changes: Raised beds built 12" high over rocky Ozark soil solve the shallow-depth problem entirely — root crops (carrots, parsnips, potatoes, beets) now thrive without fighting rock. Deep (2:1:1 topsoil/compost/coarse sand) mix provides excellent drainage for disease reduction. A hoop house or cold frame over the northernmost bed extends the growing season 4–6 weeks on each end — this is the single highest-value improvement for NW Arkansas.

New crops enabled: Parsnips (excellent caloric root for cold storage), deeper potato production, better carrot yields, late fall extension into December.

Northwest Arkansas — Caloric Output Estimate

Crop	Area (sq ft)	Est. Yield	Calories
Potato	100 sq ft	80–100 lbs	29,000–36,000
Sweet Potato	120 sq ft	75–90 lbs	31,000–38,000
Dry Beans/Cowpeas	60 sq ft	8–12 lbs dry	12,000–18,000
Winter Squash	80 sq ft	60–80 lbs	8,000–10,000
Corn (dent, dried)	80 sq ft	15–20 lbs	25,000–33,000
Tomatoes	64 sq ft	50–70 lbs	7,500–10,500
Collards/Kale	48 sq ft	25–40 lbs	4,000–6,000

Root crops (carrot, beet, turnip)	48 sq ft	30–45 lbs	4,000–6,000
TOTAL (approx)	600 sq ft active		~120,000–157,000 cal/season

NW Arkansas's two reliable seasons (spring and fall, with a hot summer production window) allow this plot to produce approximately 150,000–200,000 calories per year — roughly 20–27% of one adult's annual caloric needs from 800 sq ft. Higher than single-season northern zones due to sweet potato production and the long warm season for corn drying.

SOUTHEASTERN WISCONSIN

Milwaukee / Racine / Kenosha Area · Zone 5b–6a · ~170 Frost-Free Days

Southeastern Wisconsin Climate Profile

The southeast corner of Wisconsin is the state's best growing zone — moderated by Lake Michigan's thermal mass, with rich glacial soils and a growing season long enough for the full range of temperate vegetables. The trade-off is disease pressure from the lake's humidity and the occasional late or early frost that catches unprepared gardeners.

Factor	Data
Zone	5b–6a (lakeside 6a; inland 5b)
Last Spring Frost	April 20 – May 1
First Fall Frost	October 10–20
Frost-Free Days	165–175
Annual Rainfall	33–36 inches (relatively even; heaviest
Summer Avg High	80–85°F (comfortable; brief heat spells
Winter Lows	-5°F to -15°F possible
Native Soil	Glacial loam to clay loam; good fertilit
Soil pH	6.0–7.2 (slightly acidic to neutral — ex

Key regional advantages: Excellent native soil (no heavy clay like Houston, no rock like Arkansas). Long enough season for a complete Three Sisters planting. Lake Michigan moderates temperature extremes and extends fall.

Southeastern Wisconsin — Seed List

Crop	Best Variety	Days	Qty	Season	Purpose
Potato	Kennebec, Elba, German Butterball	70–90 days	5 lbs	Spring	Primary calories
Winter Squash	Waltham Butternut, Hubbard Blue, Delicata	85–100 days	1 pkt	Spring/Fall harvest	Long-storage calories

Dry Beans	Jacob's Cattle, Black Turtle, Soldier Be	85–95 days	½ lb	Summer	Storage protein
Corn (dent)	Bloody Butcher, Reid's Yellow Dent	95–110 days	¼ lb	Spring	Dried cornmeal
Corn (sweet)	Golden Bantam, Country Gentleman	75–85 days	¼ lb	Spring	Fresh calories
Tomato	Mortgage Lifter, Mountain Merit, Juliet	70–80 days	1 pkt	Spring	Vitamins, lycopene
Kale	Red Russian, Lacinato, Winterbor	50–65 days	1 pkt	Spring/Fall	Cold-hardy vitamins
Garlic	German Extra Hardy, Music, Chesnok Re	9 months	2 lbs	Plant Oct	Pest deterrent, calories
Onion	Copra (long storage), Red Zeppelin	100–115 days	1 pkt	Spring	Companion, storage
Carrot	Bolero, Danvers 126, Scarlet Nantes	70–80 days	1 pkt	Spring/Summer	Root cellar storage
Beet	Detroit Dark Red, Cylindra, Golden	55–65 days	1 pkt	Spring/Fall	Root + greens
Cabbage	Copenhagen Market, Late Flat Dutch (st	75–100 days	1 pkt	Spring/Fall	Fermented (sauerkraut), storage
Turnip / Rutabaga	Purple Top White Globe, Laurentian Rutab	55–90 days	1 pkt	Fall	Root cellar calories
Parsnip	Hollow Crown, Cobham Improved	100–120 days	1 pkt	Spring (long season)	Root cellar, sweet winter
Peas	Sugar Snap, Oregon Giant	60–70 days	¼ lb	Early spring	Spring protein
Snap Beans	Blue Lake Pole, Rattlesnake	55–65 days	¼ lb	Summer	Fresh and dry
Sunflower	Mammoth Russian	80–90 days	½ oz	Spring	Seeds, structure
Dill	Bouquet	40–50 days	1 pkt	Summer	Companion, culinary
Basil	Genovese	60 days	1 pkt	Summer	Tomato companion
Pepper	Carmen, Beaver Dam	65–75 days	1 pkt (8 plants)	Summer	Vitamins, storage

Southeastern Wisconsin — Planting Calendar

Month	Activity
February	Start onion seeds indoors. Order seeds.
March	Start tomatoes, peppers, kale indoors (8
Late March – April 1	Sow peas outdoors (frost-tolerant). Plan
April 1–15	Direct sow carrots, beets, spinach. Plan
April 20 – May 1	Last frost range. Set out transplants af
May 1–15	Transplant tomatoes, peppers. Direct sow
May 20	Plant corn in hills for Three Sisters.
June 1	Add beans to base of established corn (8
June 10	Add squash between corn hills.
June–August	Main growing season. Harvest continuousl
July 15 – Aug 1	Sow fall carrots, beets, turnips, kale f
August 15	Plant fall kale, spinach (will survive l
September	Begin harvesting winter squash when stem
October 1–15	Harvest all frost-tender crops before fi
October 15	Plant garlic for next year. Mulch heavil
November	Root cellar all stored crops. Mulch garl

Southeastern Wisconsin — System 1: Minimal Improvement (Mounded Rows)

What "minimal" means on glacial loam: SE Wisconsin's native soil is actually very workable. System 1 here simply means hilling potatoes, forming broad flat-topped ridges for root crops and corn, and using deep mulch for moisture retention. This soil responds well — yields will be good even without raised beds.

```
SE WISCONSIN — SYSTEM 1 LAYOUT (Mounded Rows, Glacial Loam)
Plot: 20' Wide × 40' Long · Each cell = 2'×2' · N at top

      +----------- 20 FEET ----------->
                  NORTH
  +----+----+----+----+----+----+----+----+----+----+
  - SU - SU - SU - SU - SU - SU - SU - SU - SU - SU - ← 2'   [SUNFLOWER row - N edge]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 4'   [POTATOES]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 6'
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- -- - -- - -- - - - - -- - -- -- - -- - - - ← 8'   PATH
  +----+----+----+----+----+----+----+----+----+----+
  - ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 10'  [TOMATO + BASIL]
  +----+----+----+----+----+----+----+----+----+----+
  - ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 12'
  +----+----+----+----+----+----+----+----+----+----+
  - ON - PP - PP - CA - CA - CA - CA - PP - PP - ON - ← 14'  [PEPPERS + CARROTS]
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- -- - -- - -- - - - - -- - -- -- - -- - - - ← 16'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 18'  [THREE SISTERS]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 20'
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 22'
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- -- - -- - -- - - - - -- - -- -- - -- - - - ← 24'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - KL - KL - BT - BT - BT - BT - KL - KL - GA - ← 26'  [KALE + BEETS]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - KL - KL - BT - BT - BT - BT - KL - KL - GA - ← 28'
  +----+----+----+----+----+----+----+----+----+----+
  - GA - DB - DB - DB - DB - DB - DB - DB - DB - GA - ← 30'  [DRY BEANS]
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- -- - -- - -- - - - - -- - -- -- - -- - - - ← 32'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - CA - CA - PN - PN - PN - PN - CA - CA - GA - ← 34'  [CARROTS + PARSNIPS]
```

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+----+----+----+----+----+----+----+----+----+----+
- GA - CA - CA - PN - PN - PN - PN - CA - CA - GA - ← 36'
+----+----+----+----+----+----+----+----+----+----+
- GA - TN - TN - RD - TN - TN - RD - TN - TN - GA - ← 38'  [TURNIP/RUTABAGA + RADISH]
+----+----+----+----+----+----+----+----+----+----+
- GA - TN - TN - RD - TN - TN - RD - TN - TN - GA - ← 40'
+----+----+----+----+----+----+----+----+----+----+
                    SOUTH

LEGEND: PN=Parsnip, all others as standard key.
FALL SWAP: After potatoes harvested (Aug), replant Rows 4-6 with fall kale, spinach, cabbage.
After dry beans harvested (Sept), replant Row 30 with overwintering garlic (Oct planting).
```

Southeastern Wisconsin — System 2: Full Improvement

What changes: Three 4' permanent raised beds running north-south. Primary addition: a cold frame or low hoop tunnel on the northernmost bed extends kale, spinach, and carrots into November and December — a significant caloric and nutritional extension in this zone. Garlic planted in the cold frame overwinters better and produces larger bulbs.

Season extension math: A cold frame or low tunnel on a 4' × 20' bed adds approximately 4–6 weeks of growing on each end. In Zone 5b, this means starting in late March instead of May (gaining 6 weeks of spring), and harvesting into late November instead of October (gaining 6 weeks of fall). For kale, spinach, arugula, and carrots, this is the difference between a seasonal crop and an 8–9 month harvest.

Southeastern Wisconsin — Caloric Output Estimate

Crop	Area (sq ft)	Est. Yield	Calories
Potato	120 sq ft	90–120 lbs	33,000–44,000
Winter Squash	80 sq ft	60–80 lbs	8,000–10,000
Dry Beans	60 sq ft	8–10 lbs dry	12,000–15,000
Corn (dent + sweet)	80 sq ft	15–20 lbs dried + 20 lbs fresh	28,000–36,000
Cabbage	40 sq ft	40–60 lbs	4,000–6,000
Tomatoes	48 sq ft	40–60 lbs	6,000–9,000
Root crops (carrot, beet, turnip, rutaba	80 sq ft	60–80 lbs	8,000–11,000
Kale/Collards	40 sq ft	20–30 lbs	3,000–5,000
Snap/Dry Beans (secondary)	32 sq ft	5–8 lbs dry	7,500–12,000
TOTAL (approx)	580 sq ft active		~110,000–148,000 cal/season

SE Wisconsin's single long season (170 frost-free days) produces one main harvest cycle plus a fall extension for cold-hardy crops. Approximately 110,000–148,000 calories per year — roughly 15–20% of one adult's annual caloric needs from 800 sq ft. Root cellar storage of potato, squash, cabbage, carrot, parsnip, and rutabaga extends food availability through January-February.

CENTRAL WISCONSIN

Wausau / Stevens Point Area · Zone 4b–5a · ~140 Frost-Free Days

Central Wisconsin Climate Profile

Central Wisconsin is a harder climate than the southeast — 30 fewer frost-free days, colder winters (-20°F possible), and the famous central Wisconsin sandy soil (the "sand counties" of Aldo Leopold's essays). The sand drains fast, warms quickly in spring, and is forgiving to work — but it holds neither water nor nutrients without significant amendment. The shorter season requires prioritizing fast-maturing varieties and cold-hardy crops.

Factor	Data
Zone	4b–5a
Last Spring Frost	May 1–15 (later than SE WI)
First Fall Frost	October 1–10 (earlier than SE WI)
Frost-Free Days	135–150
Annual Rainfall	30–33 inches
Summer Avg High	78–82°F
Winter Lows	-15°F to -25°F
Native Soil	Sandy loam to loamy sand (central); some
Soil pH	5.5–6.5 (often acidic sandy soil)

Key challenge: 140 days is enough for most vegetables but requires using fast-maturing varieties and timing precisely. Three Sisters works here but requires the longest-season corn varieties planted at the earliest possible date. Sandy soil means irrigation or frequent watering is critical — it dries out in 2–3 days without rain.

Central Wisconsin — Seed List

Note: All varieties selected for shorter days to maturity to fit the 140-day window

Crop	Best Variety	Days	Qty	Notes
Potato	Kennebec (75d), Norland Red (70d)	70–80 days	5 lbs	Fast-maturing varieties
Winter Squash	Delicata (90d), Acorn Table Queen (80d)	80–95 days	1 pkt	Shorter-season squash
Dry Beans	Jacob's Cattle (85d), Vermont Cranberry	80–90 days	½ lb	Push the season
Corn	Bloody Butcher starts risky—use Peache	70–95 days	¼ lb	Earliest possible planting
Tomato	Siletz (52d from transplant!), Stupice,	55–70 days	1 pkt	Cold-tolerant, fast
Kale	Red Russian (50d), Siberian (50d)	50–55 days	1 pkt	Cold-hardiest varieties
Garlic	German Extra Hardy, Music	9 months	2 lbs	Hardneck only for this zone
Peas	Maestro (61d), Sugar Ann (56d)	56–65 days	¼ lb	Critical spring crop

Carrot	Nantes (68d), Napoli (58d)	58–68 days	1 pkt	Shorter-season types
Beet	Early Wonder (55d), Red Ace (53d)	53–60 days	1 pkt	Fastest varieties
Turnip	Tokyo Market (35d), Just Right (60d)	35–60 days	1 pkt	Fastest useful root crop
Kohlrabi	Eder (38d), Grand Duke (50d)	38–50 days	1 pkt	Fastest brassica for tight windows; di
Cabbage	Savoy Ace (80d), Copenhagen Market (72d)	72–85 days	1 pkt	Stores well, cold-hardy
Snap Beans	Bush Blue Lake 274 (58d), Provider (50d)	50–58 days	¼ lb	Fast, productive
Rutabaga	Laurentian (90d)	90 days — plant early May	1 pkt	High-calorie root cellar
Sunflower	Mammoth Russian	80 days	½ oz	Needs full season
Spinach	Bloomsdale (48d), Space (40d)	40–50 days	1 pkt	Spring and fall
Basil	Genovese	60 days	1 pkt	(push season)
Dill	Bouquet (40d)	40 days	1 pkt	Easy
Onion	Yellow Sweet Spanish, Candy	100 days from set	sets or transplants	Start indoors Feb

Central Wisconsin — Planting Calendar

Month	Activity
February	Start onions, kale, early tomatoes indoo
March	Start tomato and pepper transplants indo
April 15	Direct sow peas, spinach, kale (frost-to
May 1	Plant potatoes (can tolerate light frost
May 10–15	LAST FROST window. Harden off and transp
May 15–20	Direct sow beans, corn, squash, cucumber
May 25	Three Sisters: plant corn.
June 10	Add beans to corn hills (corn should be
June 20	Add squash between hills.
July 1 – Aug 15	Main growing season. Irrigation critical
July 15 – Aug 1	Sow fall beets, turnips, kale, spinach,
September	Harvest season. Begin curing winter squa
October 1–5	Hard frost imminent — harvest everything
October 10–15	Plant garlic. Mulch heavily with 6" stra
November	Root cellar storage. Final garlic mulchi

Central Wisconsin — System 1: Minimal Improvement (Sandy Soil)

What "minimal" means on sandy soil: Unlike clay, sandy soil is easy to work but holds nothing — not water, not nutrients. System 1 in central Wisconsin requires deep organic mulch (4–6 inches of straw or wood chips) to retain the moisture the sand won't hold. Form broad, flat-topped raised rows (not mounds — mounds dry out faster in sand). Plant intensively to shade the soil surface.

Water management: Sandy soil without irrigation dries in 2–3 days without rain. In System 1, a mulched keyhole watering system (watering at the base of every third plant) is the minimum viable strategy. System 2 adds drip irrigation.

```
CENTRAL WISCONSIN — SYSTEM 1 LAYOUT (Broad Rows, Sandy Soil)
Plot: 20' Wide × 40' Long · Each cell = 2'×2' · N at top

     +----------- 20 FEET ----------->
              NORTH
  +----+----+----+----+----+----+----+----+----+----+
  - SU - SU - SU - SU - SU - SU - SU - SU - SU - SU - ← 2'   [SUNFLOWERS]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 4'   [POTATOES]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 6'
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 8'   PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 10'  [THREE SISTERS]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 12'
  +----+----+----+----+----+----+----+----+----+----+
  - GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 14'
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 16'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 18'  [TOMATO + BASIL]
  +----+----+----+----+----+----+----+----+----+----+
  - ON - TM - TM - HB - PP - PP - HB - TM - TM - ON - ← 20'  [TOMATO + PEPPER center]
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 22'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - DB - DB - DB - DB - DB - DB - DB - DB - GA - ← 24'  [DRY BEANS]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - DB - DB - DB - DB - DB - DB - DB - DB - GA - ← 26'
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 28'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - KL - KL - BT - BT - BT - BT - KL - KL - GA - ← 30'  [KALE + BEETS]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - KL - KL - CA - CA - CA - CA - KL - KL - GA - ← 32'  [KALE + CARROTS]
  +----+----+----+----+----+----+----+----+----+----+
  - -- - -- - -- - -- - -- - -- - -- - -- - -- - -- - ← 34'  PATH
  +----+----+----+----+----+----+----+----+----+----+
  - GA - TN - TN - TN - PZ - PZ - TN - TN - TN - GA - ← 36'  [TURNIPS + PEAS (spring)]
  +----+----+----+----+----+----+----+----+----+----+
  - GA - TN - TN - TN - RD - RD - TN - TN - TN - GA - ← 38'
  +----+----+----+----+----+----+----+----+----+----+
  - GA - TN - RU - RU - RU - RU - RU - RU - TN - GA - ← 40'  [RUTABAGA - high calorie root]
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+----+----+----+----+----+----+----+----+----+----+
              SOUTH


LEGEND: RU=Rutabaga; all others as standard key.
NOTE: In sandy soil, mulch ALL planted rows 4-6 inches with straw immediately after planting.
This is non-negotiable for water retention in central WI sand.
```

Central Wisconsin — System 2: Full Improvement

The single most impactful improvement for Central Wisconsin: Compost amendment of sandy soil, combined with drip irrigation. Sandy soil amended with 4–6 inches of compost incorporated 12" deep becomes a high-performing growing medium. The compost holds water and nutrients that sand cannot. Raised beds here should be at least 12" high with a 50/50 native sand and compost mix. A row cover or low tunnel on the tomato bed adds 2–3 weeks of season at each end — enough to matter significantly in Zone 4b.

Central Wisconsin — Caloric Output Estimate

Crop	Area (sq ft)	Est. Yield	Calories
Potato	120 sq ft	80–100 lbs	29,000–36,000
Winter Squash	60 sq ft	40–55 lbs	5,000–7,000
Dry Beans	48 sq ft	6–8 lbs dry	9,000–12,000
Corn (dent/sweet)	64 sq ft	10–15 lbs dried + 15 lbs fresh	20,000–27,000
Cabbage	40 sq ft	35–50 lbs	3,500–5,000
Tomatoes	40 sq ft	30–45 lbs	4,500–6,750
Root crops (carrot, beet, turnip, rutaba	60 sq ft	40–60 lbs	5,500–8,000
Kale/Spinach	40 sq ft	15–25 lbs	2,500–4,000
Peas/Snap Beans	48 sq ft	5–8 lbs dry	7,500–12,000
TOTAL (approx)	520 sq ft active		~87,000–118,000 cal/season

Central Wisconsin's short season (140 frost-free days) and sandy soil limit per-crop yields compared to the SE corner. Approximately 87,000–118,000 calories per year — roughly 12–16% of one adult's annual caloric needs from 800 sq ft. Every calorie stored (potato, squash, dry beans, root cellar crops) matters more here than in longer-season zones. System 2's season extension and soil amendment can push this 20–30% higher.

CENTRAL MICHIGAN

Lansing / Mt. Pleasant Area · Zone 5b · ~160 Frost-Free Days

Central Michigan Climate Profile

Central Michigan sits between the lake effect zones of Lake Michigan (west) and Lake Huron (east). The result is a climate somewhere between SE Wisconsin and NW Arkansas — a reliable 160 frost-free days, good glacial soils, and a Midwest-standard growing calendar. The primary challenges are spring wetness (glacial clay soils in some areas), disease pressure from Michigan's humid summers, and the occasional late spring frost that catches transplants out early.

Factor	Data
Zone	5b
Last Spring Frost	May 1–10
First Fall Frost	October 10–20
Frost-Free Days	155–165
Annual Rainfall	28–33 inches (relatively even distributi
Summer Avg High	80–86°F
Winter Lows	-5°F to -20°F
Native Soil	Glacial till — varies from sandy loam (s
Soil pH	6.0–7.0 (generally good for vegetables)

Michigan-specific notes: Septoria leaf spot and early blight on tomatoes are endemic here due to summer humidity. Disease-resistant tomato varieties are more important in Michigan than in drier climates. Soybeans (edamame and dry) grow exceptionally well in Michigan's climate — a high-protein crop worth including that the other regions don't prioritize as highly.

Central Michigan — Seed List

Crop	Best Variety	Days	Qty	Notes
Potato	Kennebec, Elba; MSU-bred: Purdue Chipper,	75–90 days	5 lbs	Disease resistance critical in humid M
Winter Squash	Butternut Waltham, Hubbard Blue, Sweet M	85–100 days	1 pkt	Excellent storage
Dry Beans / Soybean	Jacob's Cattle, Black Turtle; Envy (edam	85–95 days	½ lb	Michigan grows both well
Corn (dent)	Reid's Yellow Dent, Bloody Butcher	95–110 days	¼ lb	
Corn (sweet)	Golden Bantam, Peaches & Cream	70–80 days	¼ lb	
Tomato	Defiant PHR (late blight + Septoria resi	72–80 days	1 pkt	Disease resistance critical
Kale	Lacinato, Winterbor, Red Russian	50–65 days	1 pkt	
Garlic	Music, Chesnok Red, Georgian Crystal	9 months	2 lbs	
Onion	Copra, Red Baron	100–110 days	1 pkt or sets	
Carrot	Bolero, Scarlet Nantes	70–80 days	1 pkt	
Beet	Detroit Dark Red, Cylindra	55–65 days	1 pkt	
Cabbage	Late Flat Dutch (100d, storage), Savoy A	85–100 days	1 pkt	
Parsnip	Hollow Crown	100–120 days	1 pkt	Start early
Peas	Sugar Snap, Oregon Giant	60–70 days	¼ lb	
Snap Beans	Blue Lake Pole, Provider	55–65 days	¼ lb	

Rutabaga	Laurentian (90d)	90 days	1 pkt	Great storage root
Turnip	Purple Top (55d)	55 days	1 pkt	
Pepper	Carmen, Beaver Dam	65–75 days	1 pkt	
Sunflower	Mammoth Russian	80–90 days	½ oz	
Basil	Genovese	60 days	1 pkt	
Dill	Bouquet	40 days	1 pkt	

Central Michigan — Planting Calendar

Month	Activity
February	Start onions, kale indoors.
March	Start tomatoes, peppers, parsnips (direc
April 1	Direct sow peas, spinach, kale. Sow pars
April 15	Plant potatoes. Sow carrots, beets.
May 1–10	Last frost zone. Set out cold-hardened t
May 10–20	Direct sow beans, corn, squash, cucumber
May 25	Three Sisters: plant corn.
June 10	Add beans to Three Sisters corn.
June 20	Add squash.
June–August	Prime season. Watch for disease on tomat
July 15 – Aug 1	Sow fall kale, spinach, turnips, beets,
September	Harvest, cure squash. Dig potatoes.
October 10	First frost. Final harvest. Dig root veg
October 15–20	Plant garlic. Deep straw mulch.

Central Michigan — System 1: Minimal Improvement (Mounded Rows)

Soil note: Michigan's glacial soils are highly variable. In Ingham County (Lansing area), you may have clay loam that benefits from mounding for drainage. In Gratiot/Isabella County (Mt. Pleasant area), soils are often sandier and more workable. Adapt: mound higher in heavier soils, focus more on moisture retention in sandier soils.

```
CENTRAL MICHIGAN — SYSTEM 1 LAYOUT (Mounded Rows)
Plot: 20' Wide × 40' Long · Each cell = 2'×2' · N at top


      +------------ 20 FEET ------------>
                NORTH
  +----+----+----+----+----+----+----+----+----+----+
  - SU - SU - SU - SU - SU - SU - SU - SU - SU - SU - ← 2'   [SUNFLOWERS - N windbreak]
  +----+----+----+----+----+----+----+----+----+----+
```

```
- GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 4'   [POTATOES]
+----+----+----+----+----+----+----+----+----+----+
- GA - PT - PT - PT - PT - PT - PT - PT - PT - GA - ← 6'
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - -- - -- - -- - - - -- - ← 8'   PATH
+----+----+----+----+----+----+----+----+----+----+
- ON - TM - TM - HB - TM - TM - HB - TM - TM - ON - ← 10'  [TOMATO + BASIL]
+----+----+----+----+----+----+----+----+----+----+
- ON - TM - TM - HB - PP - PP - HB - TM - TM - ON - ← 12'
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - -- - -- - -- - - - -- - ← 14'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 16'  [THREE SISTERS - larger block]
+----+----+----+----+----+----+----+----+----+----+
- GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 18'
+----+----+----+----+----+----+----+----+----+----+
- GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 20'
+----+----+----+----+----+----+----+----+----+----+
- GA - 3S - 3S - 3S - 3S - 3S - 3S - 3S - 3S - GA - ← 22'
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - -- - -- - -- - - - -- - ← 24'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - SO - SO - SO - DB - DB - SO - SO - SO - GA - ← 26'  [SOYBEANS + DRY BEANS]
+----+----+----+----+----+----+----+----+----+----+
- GA - SO - SO - SO - DB - DB - SO - SO - SO - GA - ← 28'
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - -- - -- - -- - - - -- - ← 30'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - KL - KL - CA - CA - CA - CA - KL - KL - GA - ← 32'  [KALE + CARROTS]
+----+----+----+----+----+----+----+----+----+----+
- GA - KL - KL - BT - BT - BT - BT - KL - KL - GA - ← 34'  [KALE + BEETS]
+----+----+----+----+----+----+----+----+----+----+
- -- - -- - -- - -- - -- - - - -- - -- - -- - - - -- - ← 36'  PATH
+----+----+----+----+----+----+----+----+----+----+
- GA - PN - PN - RU - RU - RU - RU - PN - PN - GA - ← 38'  [PARSNIPS + RUTABAGA]
+----+----+----+----+----+----+----+----+----+----+
- GA - TN - TN - RD - TN - TN - RD - TN - TN - GA - ← 40'  [TURNIPS + RADISH]
+----+----+----+----+----+----+----+----+----+----+
                    SOUTH
```

LEGEND: SO=Soybean/Edamame, RU=Rutabaga, PN=Parsnip; all others as standard key.

Michigan soybean note: Start edamame for fresh eating (85 days), or let dry for storage beans.

Central Michigan — System 2: Full Improvement

What changes: Raised beds with the same soil mix as SE Wisconsin (compost amendment for clay-heavy soils, water-retention amendment for sandy soils). The key Michigan-specific addition: a low tunnel over the tomato bed using 6-mil greenhouse film extends both ends of the season and, critically, reduces foliar wetness during summer rain events — the primary driver of Septoria and early blight. Even in summer, a temporary rain shield over tomatoes dramatically reduces fungal disease load.

Central Michigan — Caloric Output Estimate

Crop	Area (sq ft)	Est. Yield	Calories
Potato	120 sq ft	90–110 lbs	33,000–40,000
Winter Squash	80 sq ft	55–75 lbs	7,000–10,000
Dry Beans/Edamame	60 sq ft	8–12 lbs dry	12,000–18,000
Corn (dent + sweet)	80 sq ft	15–20 lbs dried + 15 lbs fresh	26,000–35,000
Cabbage	40 sq ft	40–55 lbs	4,000–5,500
Tomatoes	48 sq ft	40–55 lbs	6,000–8,250
Root crops (carrot, beet, turnip, rutaba	80 sq ft	55–75 lbs	7,500–10,000
Kale	32 sq ft	15–25 lbs	2,500–4,000
Peas/Snap Beans	40 sq ft	5–8 lbs dry	7,500–12,000
TOTAL (approx)	580 sq ft active		~106,000–143,000 cal/season

Central Michigan's 160 frost-free days and heavier soils give it a production edge over Central Wisconsin. Approximately 106,000–143,000 calories per year — roughly 14–19% of one adult's annual caloric needs from 800 sq ft. Michigan's disease-resistant MSU varieties (Defiant, Huron Chipper, Missaukee) compensate for the humid summer disease pressure that reduces yields in less adapted cultivars. Root cellar storage potential is excellent.

UNIVERSAL PRINCIPLES: BOTH SYSTEMS

Seed Saving From This Garden

Every plant in these lists is selected to be either (a) open-pollinated and saveable, or (b) available in open-pollinated varieties. This matters for survival gardening: you need a seed supply that perpetuates itself.

Easiest seeds to save:

- Tomatoes: Ferment seeds in water 3 days, rinse, dry
- Beans and peas: Let pods dry completely on the plant, shell and store

- Peppers: Let fruits fully ripen to red, remove seeds, dry 2 weeks
- Squash: Scrape seeds from mature fruit, rinse, dry 3 weeks
- Kale and other brassicas: Let plants bolt in second year, collect seed pods when tan and papery
- Sunflowers: Cut head when back turns yellow, hang to dry, thresh seeds

Harder to save (requires isolation):

- Corn: Needs 1,000 feet separation from other corn varieties to stay pure — only save one variety per season
- Squash: Different species don't cross (C. pepo and C. maxima stay separate), but varieties within species will cross
- Brassicas: All broccoli, kale, cabbage, turnips, etc. cross with each other — isolate or hand-pollinate

Storage: Dry seeds thoroughly (2–3 weeks at room temperature), store in sealed glass jars in a cool, dark, dry location. Viability periods: beans/peas 3–4 years; tomatoes/peppers 4–5 years; squash 4–6 years; carrots/parsnips 2–3 years; corn 2–3 years.

Root Cellar Basics for Stored Crops

For the northern regions (Wisconsin and Michigan especially), a root cellar or cold storage is the multiplier that converts a summer garden into a year-round food supply.

Conditions by crop:

- Potatoes: 38–45°F, 85–90% humidity, dark. Do not wash before storing.
- Winter Squash: 50–55°F, 50–70% humidity. Cure at 80°F for 10 days first to harden skin.
- Carrots/Parsnips: 32–38°F, 95%+ humidity. Pack in damp sand in boxes.
- Beets/Turnips/Rutabaga: 32–40°F, 90–95% humidity. Pack in damp sand.
- Garlic: 32–40°F or 60–65°F, LOW humidity (50–60%). Never store in humid conditions.
- Onions: 32–40°F, low humidity. Cure 4 weeks with good airflow before storage.
- Dry Beans/Corn: Room temperature is fine once fully dried. Airtight container.
- Sweet Potatoes: 55–60°F, 80–90% humidity. Cure at 85°F, 90% humidity for 10 days first. Never refrigerate.

Improvised root cellar: A buried garbage can, a packed straw bale insulated earth pit, or a north-facing basement crawl space with no heating — all work for the northern regions. In Houston, sweet potatoes are stored differently: in mesh bags at 55–60°F, such as an air-conditioned interior room maintained at that temperature.

Three Sisters Guild: Detailed Planting Instructions

The Three Sisters guild appears in every regional plan. Here is the planting sequence precisely:

Step 1 (Planting Day 0): Build hills 4 feet apart, center to center. Each hill should be a flat-topped mound 6 inches high and 18 inches across. Plant 4–6 corn seeds per hill, 1 inch deep. Thin to 3 plants per hill after germination.

Step 2 (Day 14–21, when corn is 6–8 inches tall): Plant 4 bean seeds around the base of each corn hill, 6 inches from the corn stems, evenly spaced. Use climbing beans, not bush beans — pole beans need the corn for support.

Step 3 (Day 21–28): Plant 1–2 squash seeds between every pair of hills (not inside the hill). Winter squash vines will sprawl between and among the hills.

Irrigation: Water the center of each corn hill; the squash vines will find their own moisture across the bed. Do not overhead-water — ground watering only to avoid fungal issues on bean foliage.

Harvest sequence: Squash harvested when stems are dry and corky (fall). Beans left to dry on the vine (fall). Corn harvested when dry and paper-like (late fall, or left on stalk into winter if weather allows).

Area in the plan: The three rows of Three Sisters in each plan (96–128 sq ft depending on region) will produce approximately:

- Dried corn: 4–6 lbs (roughly 8,000–12,000 calories)
- Dried beans: 5–8 lbs (roughly 8,000–12,800 calories)
- Winter squash: 25–40 lbs (roughly 5,000–8,000 calories)
- Total Three Sisters block: ~21,000–32,800 calories

Rotation Schedule

Do not plant the same family in the same bed or row in consecutive years. The four main rotation families:

Family	Plants	Rotate every
Solanaceae	Tomato, pepper, eggplant, potato	3–4 years
Cucurbit	Squash, cucumber, melon	2–3 years
Legume	Beans, peas, cowpeas, soybeans	2 years
Allium	Garlic, onion, leek	2–3 years
Brassica	Kale, cabbage, turnip, radish, rutabaga	3 years
Root	Carrot, parsnip, beet	2 years

Simple four-bed rotation (System 2):

- Bed A: Solanaceae (tomatoes/peppers) → next year Legumes → year 3 Brassicas → year 4 Roots
- Bed B: Legumes → Brassicas → Roots → Solanaceae
- Bed C: Brassicas → Roots → Solanaceae → Legumes
- Bed D: Roots/Alliums → Solanaceae → Legumes → Brassicas

Three Sisters (corn/beans/squash) counts as Cucurbit + Legume combined. Rotate it out of the same ground every two years minimum.

For seed sourcing, storage guides, and additional Seedwarden products: seedwarden.com

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More from Seedwarden

If you found this guide useful, you might also like:

- Native Plants Regional Guide ($18) — Wild edible plants, trees, and fungi to supplement your survival garden by region
- Seed Saving Field Manual ($14) — Save seed from this year's harvest so your survival garden is self-renewing
- Harvest Preservation Field Manual ($16) — Can, dry, and root-cellar your surplus to last through winter and beyond

Browse all Seedwarden guides at [our Etsy shop].